# Modelling the resonant frequency behaviour for ICPT transfer using the intrinsic capacitor

Florian Copt, Jonathan Chavanne
E-mail: florian.copt@epfl.ch

***Abstract*—Inductive Coupling Power Transfer consist in a method of adding capacitors in series or parallel in order to place the circuit in resonance and transfer the power from a primary stage to a secondary stage.**

**Normally additional capacitors are added to the circuit. However it is also possible to use the intrinsic capacitance of the coil to place the circuit in resonance. While changing the geometric parameters of the coil, intrinsic capacitance and inductance will change. This paper focuses on investigating intrinsic capacitance of the coil while changing its geometry. Then resonant frequency is investigated as well.**

**The proposed study is based on a design of experiment (DOE) approach to model the different behaviours.**

## I. Introduction

More and more, implantable devices use inductive contactless power transfer (ICPT) in order to supply voltage without problems of wires. This technology consists to transfer the energy from a primary stage that contains the power supply to a secondary stage that is able to receive and convert this energy to supply necessary energy to the load. The main current example is the electrical vehicle [1], [2]. The idea is to charge the batteries located inside the frame without plugging wires to the car. The conversion chain (Fig. 1), consists in starting from DC power source that will be, convert to a high frequency signal through an inverter. A transformer without magnetic circuit is connected to the inverter. To enhance the performance of the transformer, compensation capacitors are inserted at both primary and secondary side of the transformer. As high-frequency voltage is not easily usable, a four-diode rectifier followed by a smoothing capacitor is set at the secondary side to supply the load with a DC voltage.

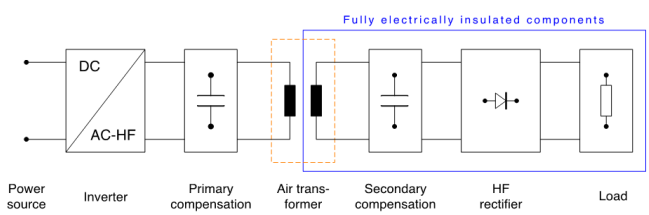



Fig. 1: Conversion chain

All the components encapsulated in the blue frame are electrically decoupled from the primary stage and all the energy is transferred through the "air transformer". The primary and secondary compensation capacitors are used to enhance the performance of the transformer and adjusted in order to work at the wished frequency. Four main well-known typologies may be considered for the compensation capacitor stages [2], [3]:

- Primary serial and secondary serial or parallel compensation (SS or SP),
- Primary parallel and secondary serial or parallel compensation (PS or PP).

Depending on the wished control (voltage control or current control), one of this topology could be chosen. For some reasons, the compensation capacitors at the second stage is not tuned in order to work at a certain frequency but the intrinsic stray capacitor ($C_2$) of the coil is used and therefore the frequency ($\omega_0$) is imposed as the resonant frequency due to this capacitor and the inductance (Ł$_2$).

$$\omega_0 = \frac{1}{\sqrt{L_2 \cdot C_2}} \tag{1}$$

The intrinsic capacitor is due to the insulation material and the air between the wires of the coil (Fig. 2). Depending on this insulation (size, material, quantity, ... ), frequency will vary a lot. Model exists but they hypothesis are contradictory. In our case, the model comes from experiments and should be more close to the reality without making assumptions. In order to determine this capacitor and therefore the resonant frequency of the coil. The problem is the fact that we ignore how the capacitor C are matched between the wires in order to create a model of the entire coil. Moreover, the parameters that influence this capacitor $C_p$ are not well determined.

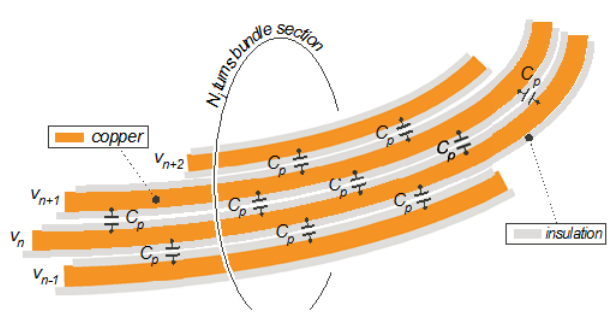



Fig. 2: Intrinsic stray capacitance

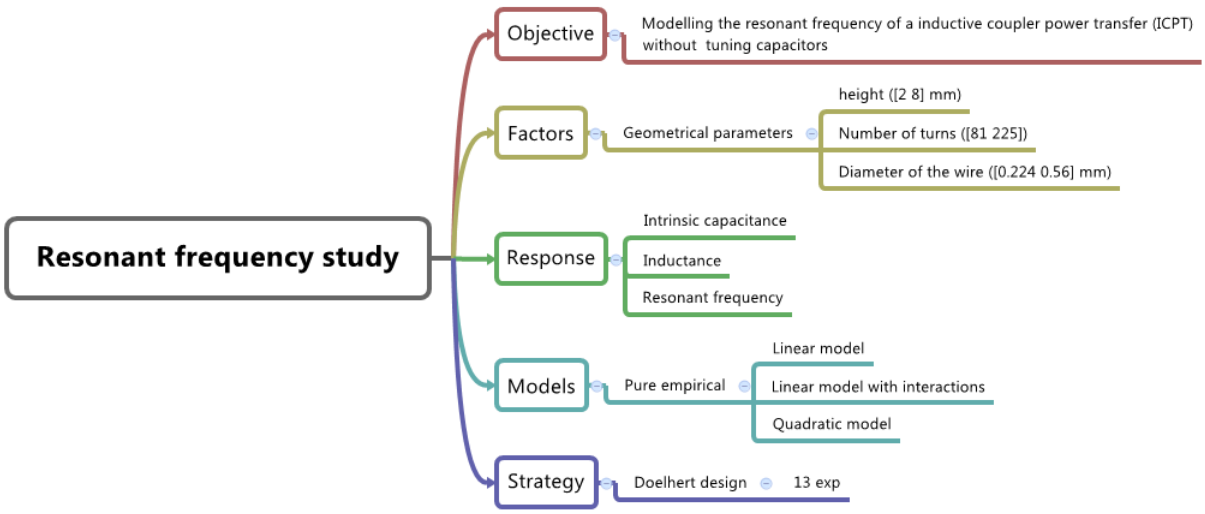



Fig. 3: Mind map of the design of experiments

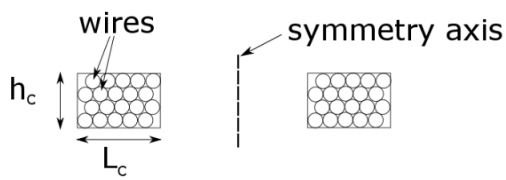



Fig. 4: Ratio parameters of the coil

## II. Design of experiments

### A. Parameters

The goal of this project is to determine and create a model that is able to specify the main parameters influencing these capacitors and so the resonant frequency. Only the second coil is investigated. Three main parameters have been chosen that could influence the frequency. The first one is the wire size. Depending on the wire diameter, the insulation will vary. A capacitor's capacitance is inversely proportional to the distance between two conductive electrodes: this parameter is certainly determinant. The insulation part represent almost 2% of the diameter of the copper. The second parameter is the number of turns ($N$) of the second coil. The frequency depends of the inductance that depends of the number of turn. The last chosen parameter permits to take into account the geometry of the coil.

The ratio between the length ($L_c$) and the height of the coil ($h_c$) (Fig. 4) will change the spreading of the coil and then may be influence the intrinsic capacitance. In order to facilitate the experiments, the ratio was not fixed but rather the height of the coil. The chosen factors are described in table I.

TABLE I: Matrix of experiments

| Factor | Parameter | Unit | Min | Max |
|---|---|---|---|---|
| Height | $X_1$ | [mm] | 2 | 10 |
| Number of turns | $X_2$ | [-] | 81 | 225 |
| Wire diameter | $X_3$ | [mm] | 0.224 | 0.56 |

### B. Models

In this design of experiment, three models (linear, linear with interaction and quadratic) have been chosen in order to describe the behaviour of the intrinsic capacitance of the coil due to the complexity to provide the main behaviour. Presence of interactions is expected between the chosen parameters and quadratic effects too. These models are therefore suitable. Their matrix are described below:

- linear:

$$Y = a_0 + \sum_{i=0}^{3} a_i X_i + \sum_{i<j}^{4} a_{ij} X_i X_j \tag{2}$$

- linear with interactions:

$$Y = a_0 + \sum_{i=0}^{3} a_i X_i \tag{3}$$

- quadratic:

$$Y = a_0 + \sum_{i=0}^{3} a_i X_i + \sum_{i<j}^{4} a_{ij} X_i X_j + \sum_{i=j}^{4} a_{ij} X_i X_j \tag{4}$$

with:

- $a_i$: coefficient containing the main effect
- $a_{ij}$: coefficient containing the interaction effect
- $Y$: response
- $X_i$ or $X_j$: variable of the model

The main problem with these three models and the chosen parameters is the cost in amount of experiments. The design of the intrinsic capacitance is not achievable by the finite element method and experiments are then necessary, the strategy is therefore determinant. As a second degree is considered, Doehlert model will be used because it requires less experiments. For three parameters, this model requires a measurement space with 13 experiments. The model matrix, which allows to explore the measurement space with 13 experiments, is given in table II.

TABLE II: Matrix of experiments

| N | 1 | $X_1$ | $X_2$ | $X_3$ |
|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 |
| 2 | 1 | -1 | 0 | 0 |
| 3 | 1 | 1 | 0 | 0 |
| 4 | 1 | -0.5 | -0.866 | 0 |
| 5 | 1 | 0.5 | 0.866 | 0 |
| 6 | 1 | -0.5 | 0.866 | 0 |
| 7 | 1 | 0.5 | -0.866 | 0 |
| 8 | 1 | -0.5 | -0.2887 | -0.8165 |
| 9 | 1 | 0.5 | 0.2887 | 0.8165 |
| 10 | 1 | -0.5 | 0.2887 | 0.8165 |
| 11 | 1 | 0 | -0.5774 | 0.8165 |
| 12 | 1 | 0.5 | -0.2887 | -0.8165 |
| 13 | 1 | 0 | 0.5774 | -0.8165 |

The Doehlert matrix of experiments is graphically represented in figure 5.

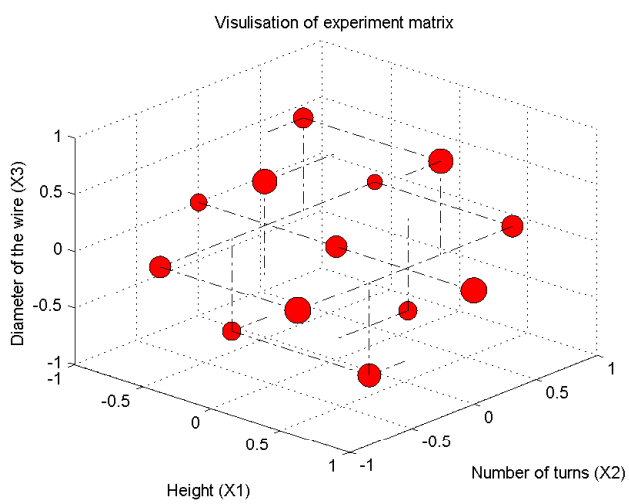



Fig. 5: Graphical representation of the matrix of experiment

## III. Results

In our study, two responses are observed: the intrinsic capacitance and the frequency response. Frequency response is given by equation 5.

$$f = \frac{1}{2\pi\sqrt{LC}} \tag{5}$$

As the main factors are the capacitance and the inductance, only each of their effects and also the response of the multiplication of these two factors will be observed.

### A. Intrinsic capacitance

*1) Model coefficient analysis:* The result of the effects for the three models (linear, linear with interaction and quadratic) are given in figure 6.

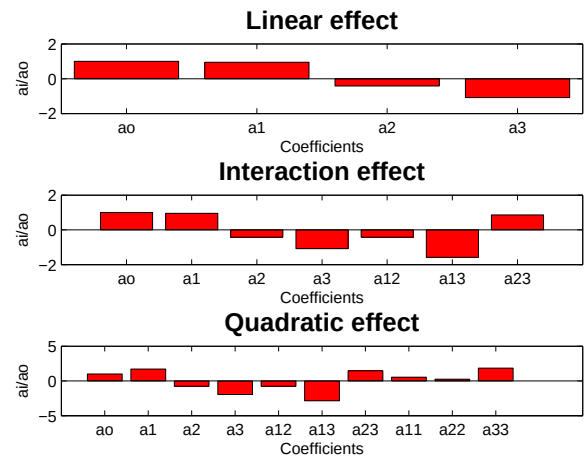



Fig. 6: Effects for the capacitance model

It is possible to notice that the height ($a_1$) and the diameter ($a_3$) are the most important parameters for the capacitance. The predominance of this effect is spread between the models. The effect of the number of turn is more negligible. By increasing the height of the coil ($a_1$), it possible to notice that the response will increase as well. Furthermore, as expected, by increasing the wire diameter ($a_3$), it increases the insulation because this last one is proportional to the wire diameter (around 2%). The underlying reason is that by increasing the diameter, the distance between the turns increases as well and as shown in equation 6, the capacitance will decrease:

$$C = \frac{\epsilon_0 \cdot \epsilon_r \cdot S}{d} \tag{6}$$

It is also possible that the strongest interaction between the main effects it that between the height and the wire diameter. For the quadratic effect only the quadratic terms of the wire size have an importance. Moreover, the other ones are negligible.

*2) ANOVA table:* The ANOVA (analysis of the variance) helps to know the accuracy of a model. For this reason, an ANOVA table on the three models is made. In our case, the results are considered as acceptable if the probability to be wrong is lower than 5%. The results are given in tables III, IV and V.

TABLE III: Anova table for linear model

| Source | SS | df | MS | F | p |
|---|---|---|---|---|---|
| Model | 1.8e-20 | 4 | 4.4e-21 | 11 | 0.0015 |
| Residual | 3.5e-21 | 9 | 3.9e-22 | | |
| Measure | 2.1e-20 | 13 | | | |

TABLE IV: Anova table for the linear model with interaction

| Source | SS | df | MS | F | p |
|---|---|---|---|---|---|
| Model | 2e-20 | 7 | 2.8e-21 | 13 | 0.0033 |
| Residual | 1.3e-21 | 6 | 2.2e-22 | | |
| Measure | 1.4e-26 | 13 | | | |

TABLE V: Anova table for the quadratic model

| Source | SS | df | MS | F | p |
|---|---|---|---|---|---|
| Model | 2e-20 | 10 | 2e-21 | 10 | 0.041 |
| Residual | 6.1e-22 | 3 | 2e-22 | | |
| Measure | 2.1e-20 | 13 | | | |

For the linear model and the linear with interaction model, the probability to be wrong ($p$) is lower than 1%. This means that the two models seem to be really accurate for our measurements. However, the quadratic model is almost at the limit of acceptance but its probability to be wrong is always acceptable.

*3) Interaction function representation:* In this section the models are represented on 3D graph where the two parameters are used (Height and Number of turns) with the response on the z-axis (figure 7, 8 and 9). In order to still have a representation of the whole set of data, three colors are used to represent the diameter parameter because the diameter parameter have 3 different values. For each value of the diameter parameter, a plan representing the model is associated.

As shown on figure 9 compared to figure 7, the experimental points fit better the quadratic function because the quadratic not only consider the interactions but also the second order relations between the points.

### B. Inductance

*1) Model coefficient analysis:* For the inductance model, it is clear on figure 10 that the main effect is the number of turns. The other evaluated parameters can be neglected compared to this effect. Also, the interaction parameter can also not be considered because there is almost no interactions between them.

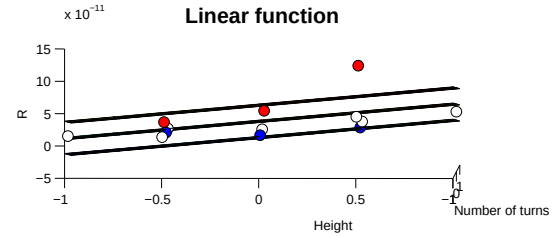



Fig. 7: Linear function for the capacitance

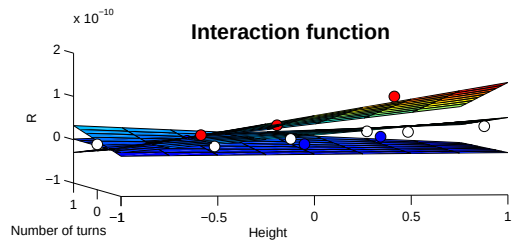



Fig. 8: Linear with interaction function for the capacitance

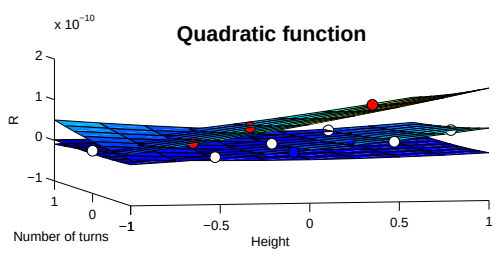



Fig. 9: Quadratic function for the capacitance

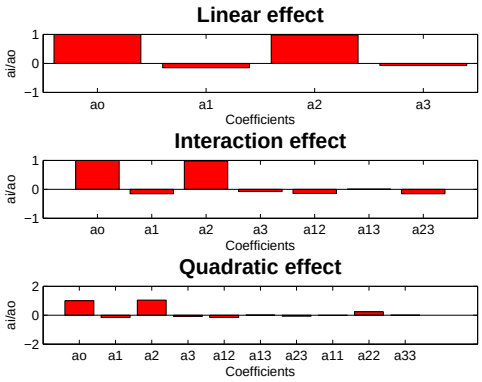



Fig. 10: Effects for the inductance

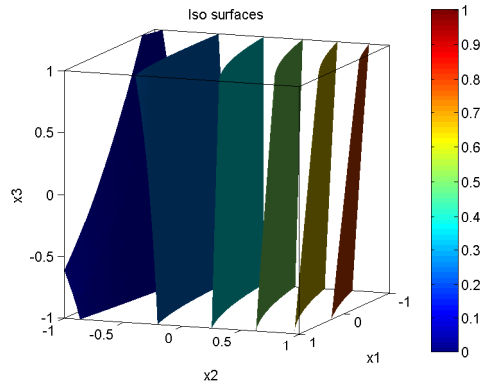



Fig. 11: ISO surface

$L$ is proportional to the section, the length of the magnetic circuit and the number of turns square while the coil is placed in a closed ferromagnetic circuit. For this reason, it is possible to expect the same behaviour in air. This justifies the dominant term ($a_2$) of our model. Nevertheless, the permanence of the air can only be calculated using finite element method (FEM).

The ANOVA tables for the three models are represented on figure VI, VII and VIII.

TABLE VI: Anova table for linear model

| Source | SS | df | MS | F | p |
|---|---|---|---|---|---|
| Model | 1.8e-5 | 4 | 4.5e-6 | 400 | 4.2e-10 |
| Residual | 1e-7 | 9 | 1.1e-8 | | |
| Measure | 1.8e-5 | 13 | | | |

TABLE VII: Anova table for the linear model with interaction

| Source | SS | df | MS | F | p |
|---|---|---|---|---|---|
| Model | 1.8e-5 | 7 | 2.6e-6 | 230 | 7.6e-7 |
| Residual | 6.7e-8 | 6 | 1.1e-8 | | |
| Measure | 1.8e-5 | 13 | | | |

TABLE VIII: Anova table for the quadratic model

| Source | SS | df | MS | F | p |
|---|---|---|---|---|---|
| Model | 1.8e-5 | 10 | 1.8e-6 | 1.5e4 | 8.4e-7 |
| Residual | 3.9e-10 | 3 | 1.2e-10 | | |
| Measure | 1.8e-5 | 13 | | | |

From the variance analysis (ANOVA), all the probability to be wrong ($p$) are smaller than $10^{-6}$. This means that all models can be chosen.

### C. Frequency

*1) Model coefficient analysis:* Unlike the intrinsic capacitance model, the response (the multiplication between the intrinsic capacitance and inductance) increases while the number of turns $a_2$ increases as well. As the relation between $L$ and $C$ is proportional to the inverse to the frequency, this last one will decrease. This means that the effect of the inductance (on the number of turns parameter) is more important than the one of the capacitance. Figure 12 shows the effect of the parameters.

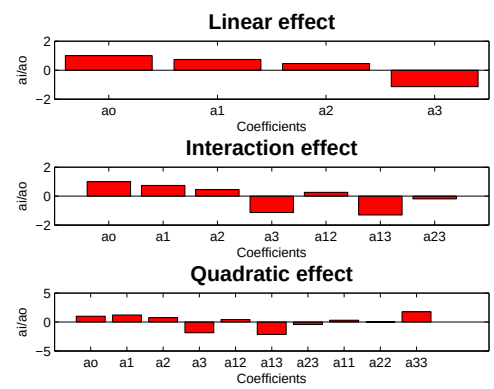



Fig. 12: Effects for the capacitance and inductance

By increasing the diameter of the wire $a_3$, the response decreases so the resonant frequency increases. For this reason, it is preferable to have a smaller wire in order to have a smaller resonant frequency.

*2) Anova table:* The variance analysis (ANOVA) for the three models are given in table IX, X and XI.

TABLE IX: Anova table for linear model

| Source | SS | df | MS | F | p |
|---|---|---|---|---|---|
| Model | 1.e-26 | 4 | 3.2e-27 | 27 | 0.00029 |
| Residual | 1.7e-27 | 9 | 1.8e-28 | | |
| Measure | 1.4e-26 | 13 | | | |

For the three models, the probability to be wrong ($p$) is lower or equal to 1%. This means that the three models seem to be really accurate for our measurements. The quadratic

TABLE X: Anova table for the linear model with interaction

| Source | SS | df | MS | F | p |
|---|---|---|---|---|---|
| Model | 1.3e-26 | 7 | 1.9e-27 | 12 | 0.004 |
| Residual | 9.8e-28 | 6 | 1.6e-28 | | |
| Measure | 1.4e-26 | 13 | | | |

TABLE XI: Anova table for the quadratic model

| Source | SS | df | MS | F | p |
|---|---|---|---|---|---|
| Model | 1.4e-26 | 10 | 1.4e-27 | 20 | 0.016 |
| Residual | 2.1e-28 | 3 | 7.1e-29 | | |
| Measure | 1.4e-26 | 13 | | | |

model is chosen for further analysis. By looking deeper into the ANOVA table of the quadratic model (fig XI) the following ANOVA table is given:

TABLE XII: Anova table details for the quadratic model

| Source | SS | df | MS | F | p |
|---|---|---|---|---|---|
| $a_0$ | 0.00 | 1 | 7.83-27 | 1.1e+2 | 0.0019 |
| $a_1$ | 0.00 | 1 | 1.3e-27 | 18 | 0.023 |
| $a_2$ | 0.00 | 1 | 4.9e-28 | 6.9 | 0.079 |
| $a_3$ | 0.00 | 1 | 3.2e-27 | 45 | 0.0068 |
| $a_{12}$ | 0.00 | 1 | 6.2e-30 | 0.087 | 0.79 |
| $a_{13}$ | 0.00 | 1 | 6.5e-28 | 9.1 | 0.057 |
| $a_{23}$ | 0.00 | 1 | 1.5e-29 | 0.21 | 0.68 |
| $a_{11}$ | 0.00 | 1 | 6.1e-29 | 0.86 | 0.42 |
| $a_{22}$ | 0.00 | 1 | 2.8e-28 | 3.9 | 0.14 |
| $a_{23}$ | 0.00 | 1 | 4.3e-28 | 6 | 0.091 |
| Residual | 0.00 | 3 | 7.1e-29 | | |

It is possible to notice that the probability to be wrong is too high for the interaction between the height and the number of turns ($a_{12}$), the interaction between the number of turns and the wire diameter ($a_{23}$), the second degree of the height ($a_{11}$) and the second degree of the number of turns ($a_{22}$). This parameters will not be taken into account for the ISO surfaces representation (figure 13). Note that the second degree effect on the coil diameter ($X_3$) is well noticeable.

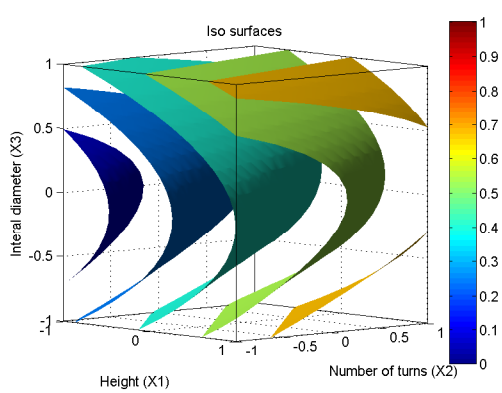
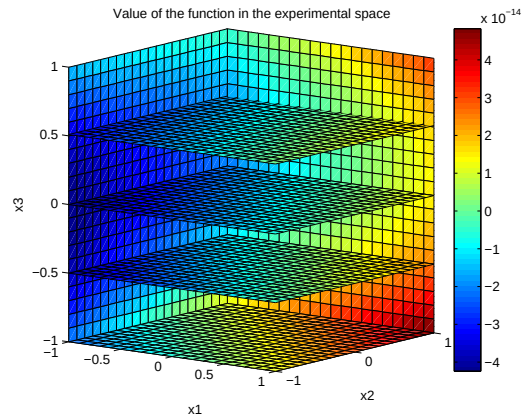



Fig. 13: ISO surfaces

In our situation, the second stage of the circuit will be placed in a limited volume. For this reason, it will be interesting to have a minimum number of turns, the minimal wire diameter and having the minimal response. As the highest $LC$ response is found with the highest height, the maximum number of turns but the smallest coil diameter, a trade-off has to be made.




## IV. Conclusion

In this paper, an experimental approach using design of experiments has been used in order to study the resonant frequency behaviour under the influence of the height of the coil, the number of turns and the wire size. An ANOVA table has been performed to ensure our results.

In order to have the smallest resonant frequency, the number of turns and the height has to be maximal but the coil diameter should be the smallest. For an application that requires a small volume, a trade-off has to be made.

By analysing the measurement and the model obtained through DOE, several hypothesis seem to appear.

For the intrinsic capacitance, we suspect that by spreading the coil turns, it seems similar to place capacitors in series. It means that the global capacitance decreases. Furthermore, by increasing the height of the coil, the global capacitance increases. This behaviour seems to act like parallel capacitor. We suspect that an important parameter was neglected which is the total length of the wire. Our hypothesis can be verified by adding this parameters in the Doehlert model.

Concerning the inductance, another parameter could be added in purpose of validating the dominance of the number of turns. Inductance being proportional to the air permanence, the internal diameter of the coil could have an main influence compared to the number of turns.

By validating this two hypothesis, it would allow us to have a better understanding over the resonant frequency.

## References


[1] S. A. et al., "Contacless power and data transfer for electric vehicle applications," in *IJAREEIE, Vol. 2, Issue 7, 2013*, vol. 2, 2013.

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